

Rajalakshmi Engineering College

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Batch: 2028
Degree: B.E - ECE

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NeoColab_REC_CS23231_DATA STRUCTURES

REC_DS using C_Week 6_PAH_Updated

Attempt : 1
Total Mark : 50
Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

You are working as a programmer at a sports academy, and the academy holds various sports competitions regularly.

As part of the academy's system, you need to sort the scores of the participants in descending order using the Quick Sort algorithm.

Write a program that takes the scores of n participants as input and uses the Quick Sort algorithm to sort the scores in descending order. Your program should display the sorted scores after the sorting process.

Input Format

The first line of input consists of an integer n, which represents the number of scores.

The second line of input consists of n integers, which represent scores separated by spaces.

Output Format

Each line of output represents an iteration of the Quick Sort algorithm, displaying the elements of the array at that iteration.

After the iterations are complete, the last line of output prints the sorted scores in descending order separated by space.

Refer to the sample outputs for the formatting specifications.

Sample Test Case

Input: 5

78 54 96 32 53

Output: Iteration 1: 78 54 96 53 32

Iteration 2: 96 54 78

Iteration 3: 78 54

Sorted Order: 96 78 54 53 32

Answer

-

Status : Skipped

Marks : 0/10

2. Problem Statement

Vishnu, a math enthusiast, is given a task to explore the magic of numbers. He has an array of positive integers, and his goal is to find the integer with the highest digit sum in the sorted array using the merge sort algorithm.

You have to assist Vishnu in implementing the merge sort algorithm.

Input Format

The first line of input consists of an integer N, representing the number of elements in the array.

The second line consists of N space-separated integers, representing the array elements.

Output Format

The first line of output prints "The sorted array is: " followed by the sorted array, separated by a space.

The second line prints "The integer with the highest digit sum is: " followed by an integer representing the highest-digit sum.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

123 456 789 321 654

Output: The sorted array is: 123 321 456 654 789

The integer with the highest digit sum is: 789

Answer

```
#include <stdio.h>
```

```
int digit_sum(int n) {  
    int sum = 0;  
    while (n) {  
        sum += n % 10;  
        n /= 10;  
    }  
    return sum;  
}
```

```
void merge(int arr[], int l, int m, int r) {  
    int n1 = m - l + 1;  
    int n2 = r - m;  
    int L[10], R[10];  
    for (int i = 0; i < n1; i++)  
        L[i] = arr[l + i];  
    for (int j = 0; j < n2; j++)  
        R[j] = arr[m + 1 + j];
```

```

int i = 0, j = 0, k = l;
while (i < n1 && j < n2) {
    if (L[i] <= R[j])
        arr[k++] = L[i++];
    else
        arr[k++] = R[j++];
}
while (i < n1)
    arr[k++] = L[i++];
while (j < n2)
    arr[k++] = R[j++];
}

```

```

void merge_sort(int arr[], int l, int r) {
    if (l < r) {
        int m = l + (r - l) / 2;
        merge_sort(arr, l, m);
        merge_sort(arr, m + 1, r);
        merge(arr, l, m, r);
    }
}

```

```

int main() {
    int n, arr[10];
    scanf("%d", &n);
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    merge_sort(arr, 0, n - 1);

    printf("The sorted array is: ");
    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);
    printf("\n");

    int max_digit_sum = -1, result = 0;
    for (int i = 0; i < n; i++) {
        int ds = digit_sum(arr[i]);
        if (ds > max_digit_sum) {
            max_digit_sum = ds;
            result = arr[i];
        }
    }
}

```

```
}  
printf("The integer with the highest digit sum is: %d\n", result);  
return 0;  
} // You are using GCC
```

Status : Correct

Marks : 10/10

3. Problem Statement

You are working on an optimization task for a sorting algorithm that uses insertion sort. Your goal is to determine the efficiency of the algorithm by counting the number of swaps needed to sort an array of integers.

Write a program that takes an array as input and calculates the number of swaps performed during the insertion sort process.

Example 1:

Input:

5

2 1 3 1 2

Output:

4

Explanation:

Step 1: [2, 1, 3, 1, 2] (No swaps)

Step 2: [1, 2, 3, 1, 2] (1 swap, element 1 shifts 1 place to the left)

Step 3: [1, 2, 3, 1, 2] (No swaps)

Step 4: [1, 1, 2, 3, 2] (2 swaps; element 1 shifts 2 places to the left)

Step 5: [1, 1, 2, 2, 3] (1 swap, element 2 shifts 1 place to the left)

Total number of swaps: $1 + 2 + 1 = 4$

Example 2:

Input:

7

12 15 1 5 6 14 11

Output:

10

Explanation:

Step 1: [12, 15, 1, 5, 6, 14, 11] (No swaps)

Step 2: [12, 15, 1, 5, 6, 14, 11] (1 swap, element 15 shifts 1 place to the left)

Step 3: [12, 15, 1, 5, 6, 14, 11] (No swaps)

Step 4: [1, 12, 15, 5, 6, 14, 11] (2 swaps, element 1 shifts 2 places to the left)

Step 5: [1, 5, 12, 15, 6, 14, 11] (1 swap, element 5 shifts 1 place to the left)

Step 6: [1, 5, 6, 12, 15, 14, 11] (2 swaps, element 6 shifts 2 places to the left)

Step 7: [1, 5, 6, 12, 14, 15, 11] (1 swap, element 14 shifts 1 place to the left)

Step 8: [1, 5, 6, 11, 12, 14, 15] (3 swaps, element 11 shifts 3 places to the left)

Total number of swaps: $1 + 2 + 1 + 2 + 1 + 3 = 10$

Input Format

The first line of input consists of an integer n , representing the number of elements in the array.

The second line of input consists of n space-separated integers, representing the elements of the array.

Output Format

The output prints the number of swaps performed during the insertion sort process.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 5

2 1 3 1 2

Output: 4

Answer

```
#include <stdio.h>
```

```
int main() {
    int n, i, j, key, swaps = 0;
    scanf("%d", &n);
    int arr[n];

    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    for (i = 1; i < n; i++) {
        key = arr[i];
        j = i - 1;
        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            swaps++;
            j--;
        }
        arr[j + 1] = key;
    }

    printf("%d\n", swaps);
    return 0;
}
```

Status : Correct

Marks : 10/10

4. Problem Statement

You're a coach managing a list of finishing times for athletes in a race. The times are stored in an array, and you need to sort this array in ascending order to determine the rankings.

You'll use the insertion sort algorithm to accomplish this.

Input Format

The first line of input contains an integer n , representing the number of athletes.

The second line contains n space-separated integers, each representing the finishing time of an athlete in seconds.

Output Format

The output prints the sorted finishing times of the athletes in ascending order.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

75 89 65 90 70

Output: 65 70 75 89 90

Answer

```
#include <stdio.h>
```

```
int main() {
    int n, i, j, key;
    scanf("%d", &n);
    int times[n];
    for(i = 0; i < n; i++) {
        scanf("%d", &times[i]);
    }
    for(i = 1; i < n; i++) {
        key = times[i];
        j = i - 1;
        while(j >= 0 && times[j] > key) {
            times[j + 1] = times[j];
```

```

        j = j - 1;
    }
    times[j + 1] = key;
}
for(i = 0; i < n; i++) {
    printf("%d ", times[i]);
}
return 0;
}

```

// You are using GCC

Status : Correct

Marks : 10/10

5. Problem Statement

Alex is working on a project that involves merging and sorting two arrays. He wants to write a program that merges two arrays, sorts the merged array in ascending order, removes duplicates, and prints the sorted array without duplicates.

Help Alex to implement the program using the merge sort algorithm.

Input Format

The first line of input consists of an integer N, representing the number of elements in the first array.

The second line consists of N integers, separated by spaces, representing the elements of the first array.

The third line consists of an integer M, representing the number of elements in the second array.

The fourth line consists of M integers, separated by spaces, representing the elements of the second array.

Output Format

The output prints space-separated integers, representing the merged and sorted array in ascending order, with duplicate elements removed.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

1 2 3 4

3

3 4 5

Output: 1 2 3 4 5

Answer

```
#include <stdio.h>
```

```
void merge(int arr[], int l, int m, int r) {  
    int n1 = m - l + 1;  
    int n2 = r - m;  
    int L[20], R[20];
```

```
    for (int i = 0; i < n1; i++)  
        L[i] = arr[l + i];  
    for (int j = 0; j < n2; j++)  
        R[j] = arr[m + 1 + j];
```

```
    int i = 0, j = 0, k = l;  
    while (i < n1 && j < n2) {  
        if (L[i] <= R[j])  
            arr[k++] = L[i++];  
        else  
            arr[k++] = R[j++];  
    }
```

```
    while (i < n1)  
        arr[k++] = L[i++];  
    while (j < n2)  
        arr[k++] = R[j++];  
}
```

```
void mergeSort(int arr[], int l, int r) {  
    if (l < r) {  
        int m = l + (r - l) / 2;
```

```

mergeSort(arr, l, m);
mergeSort(arr, m + 1, r);
merge(arr, l, m, r);
}
}

int main() {
    int N, M, a[20], b[20], merged[40], total = 0;

    scanf("%d", &N);
    for (int i = 0; i < N; i++)
        scanf("%d", &a[i]);

    scanf("%d", &M);
    for (int i = 0; i < M; i++)
        scanf("%d", &b[i]);

    for (int i = 0; i < N; i++)
        merged[total++] = a[i];
    for (int i = 0; i < M; i++)
        merged[total++] = b[i];

    mergeSort(merged, 0, total - 1);

    printf("%d", merged[0]);
    for (int i = 1; i < total; i++) {
        if (merged[i] != merged[i - 1])
            printf(" %d", merged[i]);
    }

    return 0;
} // You are using GCC

```

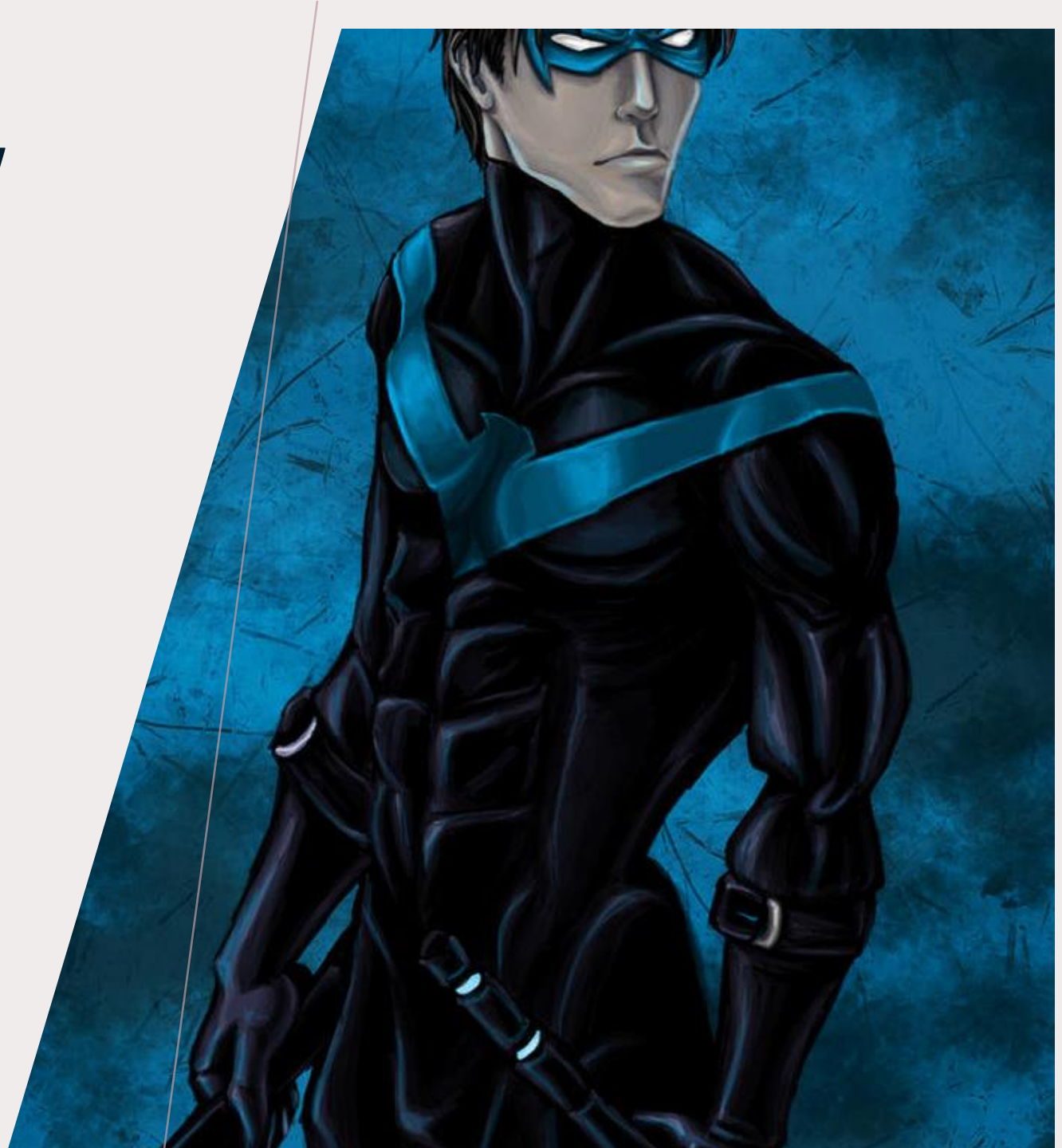
Status : Correct

Marks : 10/10

VIGILANTE
&
ANTI -
HEROES

TECHNICAL COMMUNICATION 2

~SANJAY RAM K / 240801296





INTRODUCTION:-

- A **vigilante** is a person who takes the law into their own hands without legal authority, typically because they believe that the legal system is not doing enough to deliver justice or protect society. Vigilantes often act outside the law, using force or intimidation to punish criminals or wrongdoers.

*An **anti-hero** is a central character in a story who **lacks traditional heroic qualities**, such as idealism, courage, or morality. Unlike typical heroes who fight for justice or the greater good, anti-heroes are often flawed, morally ambiguous, or even selfish—but they're still the main character or protagonist.

*EXAMPLE:- Vigilante:- Batman,daredevil,punisher,etc;

Anti-hero:-Deadpool,Walter white,Tony Montana,etc;

KEY CHARACTERISTICS OF VIGILANTE:-

- **Acts without legal authority** (not a police officer or official).
- **Motivated by a sense of justice**, revenge, or moral duty.
- **Often targets criminals** who they believe are not being properly punished.
- **Methods may be illegal or violent**, such as assault, kidnapping, or even killing.
- **Never shows mercy** , and always seeks justice .



KEY CHARACTERISTICS OF ANTI HEROES:-

- **Flawed or morally grey:** They might lie, cheat, or kill, but often for what they believe is a good reason.
- **Self-interested or cynical:** They may not believe in noble causes, but still end up doing the right thing—intentionally or by accident.
- **Relatable:** Audiences often connect with anti-heroes because they seem more realistic and human.
- **Complex motives:** They may want revenge, personal gain, or simply to survive—not to save the world.



DIFFERENCES:-

Trait

Anti-Hero

Vigilante

Morality

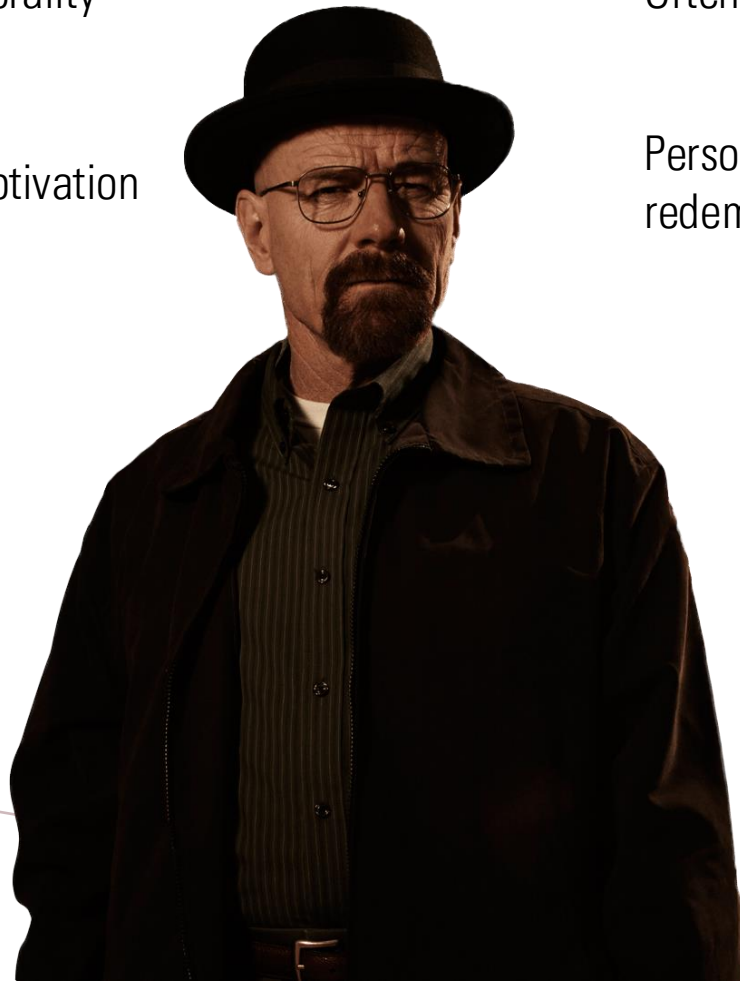
Often morally gray or self-serving

Often believes they're enforcing justice

Motivation

Personal goals, survival, revenge, redemption

Fixing justice system failures, punishing wrongdoers





BATMAN:-

- Batman is known as a **vigilante** because he **deliberately operates outside the law** to fight crime in his own way. Here's a clear breakdown of **why he's labeled a vigilante**:
- Batman is not a police officer, detective, or government agent.
- He wears a mask and uses secret identities and illegal methods to fight crime.
- He interrogates people using fear or violence.
- He makes his **own decisions** about guilt and punishment.
- Gotham is portrayed as corrupt and broken.
- Batman acts because he believes the **justice system can't handle the real crime**.
- He refuses to kill, but he has **no issue breaking laws** to stop crime.
- He uses high-tech surveillance, advanced weapons, and fear tactics—**methods that are illegal in real life**.

GHOST RIDER:-

- **Ghost Rider** is considered an **anti-hero** because he combines heroic intentions with **dark, violent, and morally extreme actions**. His powers come from **hellish or supernatural sources**, and his methods are **brutal**, often involving torture, death, and revenge.
- A human **possessed by a Spirit of Vengeance**—a demon that punishes the wicked.
- Transforms into a **flaming-skulled skeleton** with hellfire abilities, a motorcycle, and a chain weapon.
- He doesn't arrest or rehabilitate criminals—he **destroys them**.
- His punishment can be **disproportionate** and ruthless.
- His primary motive is **revenge**—either for himself or others.
- He only targets people he deems guilty, not always by law but by **moral or supernatural judgment**.



CONCLUSION:-

- A **vigilante** is defined by their **actions**: someone who takes the law into their own hands to fight injustice, often because they believe the legal system is failing. Their focus is usually on **justice or punishment**, even if it means breaking the law.
- An **anti-hero** is defined by their **character traits**: a flawed protagonist who lacks the typical virtues of a hero—like kindness, morality, or selflessness—but is still the main figure of the story. They often do the right thing, but for the **wrong reasons**, or in morally questionable ways.